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Implementation of Apriori Algorithm for Analysis of Consumer Purchase Patterns

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Abstract. Consumer purchasing patterns are a form of purchases made by consumers, whether someone or a lot of people to get the desired item by making a purchase transaction. One characteristic of the purchase pattern is the existence of acquiring something through exchanging money. This study aims to create an application that is used in determining consumer purchasing patterns by applying a priori algorithms and using Visual Basic 2010 as a tool for determining consumer purchase patterns. This application uses a priori algorithm calculation method where the sample consumer purchase data will be sorted and calculated by providing the value of the minium support and configuration parameters and based on the results of confidence the largest number of conclusions such as: can be used as information for determining sales, the application of a priori algorithms can provide information pattern combination item set from consumer purchase data that is with support above 15% and confidence above 50% on item set.

1. Introducing

Cafe Bojack Coffee Shop is a business entity that is engaged in selling light dishes for the general public. where is this cafe one of hundreds of cafes in Medan City. Although the cafe is very crowded, but often experience problems such as not the availability of the consumer order menu even though the order is in demand. Basically, the Cafe owner of the Bojack Coffe Shop has not analyzed the data specifically for example to combine items which will cause problems so that they do not know how the relationship between items with other items. Data mining is the process of finding interesting patterns or information in selected data using certain techniques or methods. The techniques, methods, or algorithms in data mining vary greatly. The selection of the right method or algorithm depends very much on the objectives and the Knowledge Discovery in Database (KDD) process in its entirety. Data mining techniques to find associative rules or relationships between items are called association rule mining. one of the algorithms used to find association rules is a priori algorithm. The Apriori Algorithm, helps in forming possible combination item candidates, then tests whether the combination meets the minimum support parameters and minimum confidence which is the threshold value given by the user.

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2. Research Methodology

2.1. Data Mining

Data mining or often referred to as knowledge discovery in databases (KDD) is an activity that includes gathering, using historical data to find order, patterns or relationships in large data. This data mining output can be used to help make future decisions. The development of KDD has caused the use of pattern recognition to diminish because it has become a part of data mining [4].

2.2. Association Rule

Following are the steps of association rules calculation process [3].

a. The system scans the database to get a 1-itemet candidate (set of items consisting of 1 item) and calculates the support value. Then the support value is compared with the specified minimum support, if the value is greater or equal to the minimum support then the itemset is included in a large itemset.
b. Itemset which is not included in large itemsets is not included in the next iteration in pruning.
c. In the second iteration the system will use large itemset results in the first iteration (L1) to form the second itemset candidate (L2). In the next iteration, the system will use the results of large itemset in the iteration and then will use large itemset results in the previous iteration (Lk-1) to form the following itemset candidate (Lk). The system will join Lk-1 with Lk-1 to get Lk, as in the previous iteration the system will delete / prune the itemset combination which is not included in the large itemset.
d. After the join operation, the new itemset result from the join process is calculated for support.
e. The candidate formation process which consists of a join and prune process will continue to be carried out until the candidate itemset set is null, or there are no more candidates to be formed.
f. After that, the result of frequent itemset was formed an association rule that met the specified support and confidence values.
g. In the formation of association rule, the same value is considered as one value.
h. The association rule that is formed must meet the specified minimum value.

Minimal Support: a value determined by the researcher to cut the combination of set items into fewer [2]. Minimal Confidence is a value that is also determined by the researcher to cut the combination of each k-item set (the result of minimal support trimming) to form association rules [2].
The basic methodology of association analysis is divided into two stages:

- *Support*
  Support from an association rule is the presentation of the combination of items in the database, where if have item A and item B then the support is the proportion of transactions in the database containing A and B. [7].
  The support value of an item is obtained by the formula[6].

$$Support(A) = \frac{(The\ number\ of\ transactions\ containing\ A)}{Transaction\ Total}$$

  While the support value of 2 items is obtained from the following formula:

$$Support(A, B) = P(A \cap B)$$

$$Support(A, B) = \frac{\sum Transactions\ contain\ A\ and\ B}{\sum Transaction}$$

- *Confidence*
  Confidence of association rule is a measure of the accuracy of a rule, which is the presentation of a transaction in a database containing A and containing B [7].

$$Confidence = P(B|A) = \frac{\sum Transactions\ contain\ A\ and\ B}{\sum Transactions\ contain\ A}$$

2.3. Apriori Algorithm

Broadly speaking the a priori algorithm works [1].

a. The formation of candidate itemset, candidate k-itemset is formed from a combination of item (k-1) -itemset obtained from the previous iteration. One characteristic of the Apriori algorithm is the presence of candidate k-itemset trimming whose subset containing k-1 items is not included in the high frequency pattern with a length of k-1.
b. Calculation of support from each candidate k-itemset. Support from each of our candidate candidates is obtained by scanning the database to calculate the number of transactions that contain all items in the candidate items. This is also a feature of the Apriori algorithm which requires calculation by scanning the entire database as much as the longest item-itemet.
c. Set a high frequency pattern. A high frequency pattern that contains k items or itemset is determined from the candidate kitet whose support is greater than the minimum support.
d. If no new high frequency pattern is obtained, the whole process is stopped. If not, then k plus one and return to part 1.

3. Results and Discussion

3.1. Apriori Algorithm Analyst

a. High Frequency Pattern Analysis

This stage looks for item combinations that meet the minimum requirements of the support value in the database.

Table 1. Consumer Purchase Transaction Data

Transaction	Items purchased
1	Dark Chocolate, Sweet Tea, Fried Rice, Rice, Cow Eye Egg, Chicken Nugget
2	Sprite, Dark Chocolate, Cappuccino, Teh Tarik, Noodle soup, Fried noodles, Nutri Sari, Sweet Tea, Baked Banana
3	Fried Rice, Brown Chocolate, Dutch Eggplant, Sweet Tea, Dark Chocolate, Fried Noodle
4	Milo, Teh Tarik, Baked Banana, Fried Banana
5	Gravy Noodles, Fried Rice, Sweet Tea,Milk
6	Teh Tarik, Kuku Bima, Sweet Tea, French Fries, Fried Noodle, Fried Rice
7	Milk, Fried Rice, Lemon Tea, Cappucino, Aqua
8	Fried Rice, Lemon Tea, Milo, Kuku Bima, Fried noodles
9	Cappucino, Milo, Martabak Mie, Milk, Noodle soup
10	French Fries, Milk Tea, Fried Rice, Nutri Sari, Cappucino, Dark Chocolate
11	Sweet Tea, Cappuccino, Milo, French Fries, Fried Rice, Grilled Banana
12	Fried Noodles, Martabak Noodles, Noodles, Rice, Tarik Tea, Sweet Tea, Milk Tea
13	Noodle soup, Oranges, Rice, Cappucino
14	Nutri Sari, Cappucino, Milo, Dutch Eggplant, Fried Rice, Coffee Mix, Avocado
15	Sweet Tea, Nasi Goreng, Coca-Cola, Baked Banana
16	Nutri Sari, Teh Tarik, Cappucino, Avocado, Fried noodles, Fried Rice, Noodle soup
17	Lemon Tea, Teh Tarik, Milk, Fried Rice, Noodle soup
18	Cappuccino, Sweet Tea, Milo, Noodles, Rice, Dark Chocolate
19	Dark Chocolate, Cappuccino, Fried Rice, Noodle soup
20	Fried Rice, Teh Tarik, Cappuccino, Fried Noodle, Dark Chocolate

For example in this study the analyst wants a rule that has more than 15% support and more than 50% confidence.

Step 1. Look for K1 (1-itemset candidate) as follows.

Table 2. Candidate 1-itemset (K1)

Item	Candidate
Aqua	1/20 * 100% = 5%
Avocado	2/20 * 100% = 10%

Item	Candidate
Brown Chocolate	1/20 * 100% = 5%
Coca-Cola	1/20 * 100% = 5%
Chicken Nugget	1/20 * 100% = 5%
Cappuccino	11/20 * 100% = 55%
Dark Chocolate	7/20 * 100% = 35%
French Fries	3/20 * 100% = 15%
Kuku Bima	2/20 * 100% = 10%
Coffee Mix	1/20 * 100% = 5%
Lemon Tea	3/20 * 100% = 15%
Fried noodles	7/20 * 100% = 35%
Noodle soup	9/20 * 100% = 45%
Milo	5/20 * 100% = 30%
Martabak Noodles	2/20 * 100% = 10%
Eyes of Cattle Eggs	1/20 * 100% = 5%
Fried rice	14/20 * 100% = 70%
Rice	4/20 * 100% = 20%
Nutri Sari	4/20 * 100% = 20%
Oranges	1/20 * 100% = 5%
Grilled banana	4/20 * 100% = 20%
Fried bananas	1/20 * 100% = 5%
Sprite	2/20 * 100% = 10%
Milk	3/20 * 100% = 15%
Dutch eggplant	2/20 * 100% = 10%
Sweet tea	9/20 * 100% = 45%
Milk tea	2/20 * 100% = 10%
Teh Tarik	7/20 * 100% = 35%

The combination of itemset in f1 can be used as a 2-itemset broker. itemset-itemset from f1 that can be used are itemset-itemset which have similarities in the first k-1 item.

The 2-itemset candidate that can be formed from f1 is as follows.

Table 3. Candidate 2-itemset (K2)

No	Item 1	Item 2	Candidate
1	Fried rice	Cappuccino	7/20 * 100% = 35%
2	Fried rice	Noodle soup	4/20 * 100% = 20%
3	Fried rice	Sweet tea	6/20 * 100% = 30%
4	Fried rice	Dark Chocolate	5/20 * 100% = 25%
5	Fried rice	Teh Tarik	4/20 * 100% = 20%
6	Fried rice	Fried noodles	5/20 * 100% = 25%
7	Fried rice	Milo	3/20 * 100% = 15%
8	Fried rice	White rice	1/20 * 100% = 5%
9	Fried rice	Nutri Sari	3/20 * 100% = 15%
10	Fried rice	Grilled banana	2/20 * 100% = 10%
11	Fried rice	milk	3/20 * 100% = 15%
12	Nasi Goreng	Lemon Tea	3/20 * 100% = 15%
13	Fried rice	French Fries	3/20 * 100% = 15%
14	Cappucino	Noodle soup	6/20 * 100% = 30%
15	Cappucino	Sweet tea	3/20 * 100% = 15%
16	Cappucino	Dark Chocolate	5/20 * 100% = 25%
17	Cappucino	Teh tarik	3/20 * 100% = 15%
18	Cappucino	Fried noodles	3/20 * 100% = 15%
19	Cappucino	Milo	4/20 * 100% = 20%
20	Cappucino	rice	2/20 * 100% = 10%
21	Cappucino	Nutri Sari	4/20 * 100% = 20%
22	Cappucino	Grilled banana	2/20 * 100% = 10%

No	Item 1	Item 2	Candidate
23	Cappucino	White milk	2/20 * 100% = 10%
24	Cappucino	Lemon Tea	1/20 * 100% = 5%
25	Cappucino	French Friesh	2/20 * 100% = 10%
26	Noodle soup	Sweet tea	4/20 * 100% = 20%
27	Noodle soup	Dark Chocolate	3/20 * 100% = 15%
28	Noodle soup	Teh Tarik	4/20 * 100% = 20%
29	Noodle soup	Fried noodles	3/20 * 100% = 15%
30	Noodle soup	Milo	2/20 * 100% = 10%
31	Noodle soup	rice	3/20 * 100% = 15%
32	Noodle soup	Nutri Sari	2/20 * 100% = 10%
33	Noodle soup	Grilled banana	1/20 * 100% = 5%
34	Noodle soup	milk	3/20 * 100% = 15%
35	Noodle soup	Lemon Tea	1/20 * 100% = 5%
36	Noodle soup	French Fries	0/20 * 100% = 0%
37	Sweet tea	Dark Chocolate	4/20 * 100% = 20%
38	Sweet tea	Teh Tarik	3/20 * 100% = 15%
39	Sweet tea	Fried noodles	4/20 * 100% = 20%
40	Sweet tea	Milo	2/20 * 100% = 10%
41	Sweet tea	rice	3/20 * 100% = 15 %
42	Sweet tea	Nutri Sari	1/20 * 100% = 5%
43	Sweet tea	Grilled banana	3/20 * 100% = 15 %
44	Sweet tea	milk	1/20 * 100% = 5%
45	Sweet tea	Lemon Tea	0/20 * 100% = 0%
46	Sweet tea	Frenh Fries	2/20 * 100% = 10%
47	Dark Chocolate	Teh Tarik	2/20 * 100% =10 %
48	Dark Chocolate	Fried noodles	3/20 * 100% = 15%
49	Dark Chocolate	Milo	1/20 * 100% = 5%
50	Dark Chocolate	White rice	2/20 * 100% = 10%
51	Dark Chocolate	Nutri Sari	2/20 * 100% =10 %
52	Dark Chocolate	Grilled banana	1/20 * 100% = 5%
53	Dark Chocolate	milk	0/20 * 100% = 0 %
54	Dark Chocolate	Lemon Tea	0/20 * 100% = 0 %
55	Dark Chocolate	French Fries	1/20 * 100% = 5 %
56	Teh Tarik	Milo	1/20 * 100% =5 %
57	Teh Tarik	White rice	1/20 * 100% = 5%
58	Teh Tarik	Nutri Sari	2/20 * 100% = 10%
59	Teh Tarik	Grilled banana	2/20 * 100% = 10%
60	Teh Tarik	milk	1/20 * 100% = 5%
61	Teh Tarik	Lemon Tea	1/20 * 100% = 5%
62	Teh Tarik	French Friesh	1/20 * 100% = 5 %
63	Fried noodles	Milo	1/20 * 100% = 5 %
64	Fried noodles	rice	1/20 * 100% = 5%
65	Fried noodles	Nutri Sari	2/20 * 100% = 10%
66	Fried noodles	Grilled banana	1/20 * 100% = 5 %
67	Fried noodles	milk	0/20 * 100% = 0%
68	Fried noodles	Lemon Tea	1/20 * 100% = 5 %
69	Fried noodles	French Fries	1/20 * 100% = 5 %
70	Milo	White rice	1/20 * 100% = 5 %
71	Milo	Nutri Sari	1/20 * 100% = 5%
72	Milo	Grilled banana	2/20 * 100% = 10%
73	Milo	milk	1/20 * 100% = 5%
74	Milo	Lemon Tea	1/20 * 100% = 5 %
75	Milo	French Fries	0/20 * 100% = 0 %
76	Rice	Nutri Sari	0/20 * 100% = 0 %
77	Rice	Grilled banana	0/20 * 100% = 0%
78	Rice	milk	0/20 * 100% = 0%

No	Item 1	Item 2	Candidate
79	Rice	Lemon Tea	0/20 * 100% = 0%
80	Rice	French Fries	0/20 * 100% = 0 %
81	Nutri Sari	Grilled banana	1/20 * 100% = 5 %
82	Nutri Sari	milk	0/20 * 100% = 0%
83	Nutri Sari	Lemon Tea	0/20 * 100% = 0%
84	Nutri Sari	French Fries	1/20 * 100% = 5%
85	Milk	Lemon Tea	2/20 * 100% = 10%
86	Milk	French Fries	0/20 * 100% = 0 %
87	Lemon Tea	French Fries	0/20 * 100% = 0 %

After all high frequency patterns have been found, then the association rules that meet the minimum requirements are found for confidence associative rules A → B minimum confidence = 50%.

Table 4. Candidate association rules from F2

Itemset	Support	Confidence
If buy Fried Rice, will buy Cappuccino	7/14	50%
If buy Fried Rice, will buy Noodles soup	4/14	28,57%
If buy Fried Rice, will buy Sweet Tea	6/14	42,85%
If buy Fried Rice, will buy Chocolate	5/14	35,71%
If buy Fried Rice, will buy Teh Tarik	4/14	28,57%
If buy Fried Rice, will buy Fried Noodles	5/14	35,71%
If buy Fried Rice, will buy Nutri Sari	3/14	21,42%
If buy Fried Rice, will buy LemonTea	3/14	21,42%
If buy Fried Rice, will buy French Fries	3/14	21,42%
If buy Cappuccino, will buy Noodle soup	6/11	54,55%
If buy Cappuccino, will buy Sweet Tea	4/11	27,27%
If buy Cappuccino, will buy Dark Chocolate	5/11	45,45%
If buy Cappuccino, will buy Teh Tarik	3/11	27,27%
If buy Cappuccino, will buy Fried noodles	3/11	27,27%
If buy Cappuccino, will buy Nutri Sari	4/11	36,36%
If buy Noodle soup, will buy Sweet Tea	4/9	44,44%
If buy Noodle soup, will buy Dark Chocolate	3/9	33,33%
If buy Noodle soup, will buy Teh Tarik	4/9	44,44%
If buy Noodle soup, will buy Fried Noodles	3/9	33,33%
If buy Noodle soup, will buy Rice	3/9	33,33%
If buy Mie Kuah, will buy Milk	3/9	33,33%
If buy Sweet Tea, will buy Dark Chocolate	4/9	44,44%
If buy Manis Teh, will buy Teh Tarik	3/9	33,33%
If buy Sweet Tea, will buy Fried Noodles	4/9	44,44%
If buy Sweet Tea, will buy Rice	3/9	33,33%
If buy Dark Chocolate, will buy Fried Noodles	3/7	42,85%
If buy Teh Tarik, will buy Fried Noodles	5/7	71,43%s

b. Final Association Rules

Association rules are sequential based on minimum support and minimum confidence that meet the following:

Table 5. Candidate association rules from F2

Itemset	Support	Confidence
If you buy Fried Rice, you will buy Cappuccino	35%	50%
If you buy Capuucino, you will buy Noodles soup	30%	54,55%
If you buy Teh Tarik, you will buy Fried Noodles	25%	71,43%

c. System Design

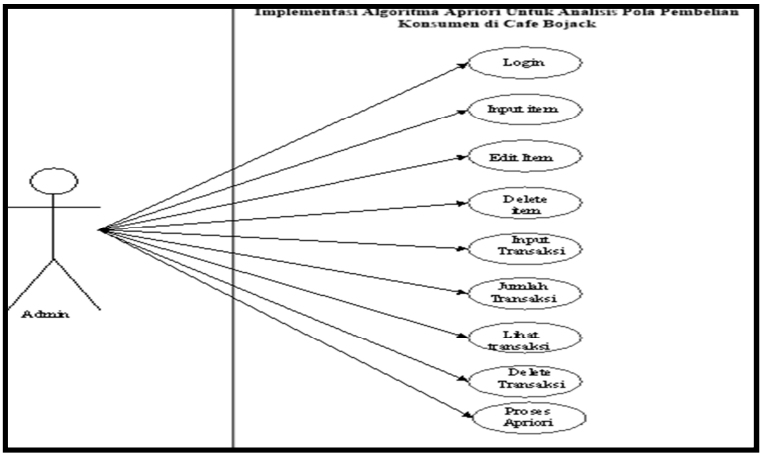

Figure 1. Use Case Diagram

3.2. Implementation System

This page is used to display the results of the data mining process.

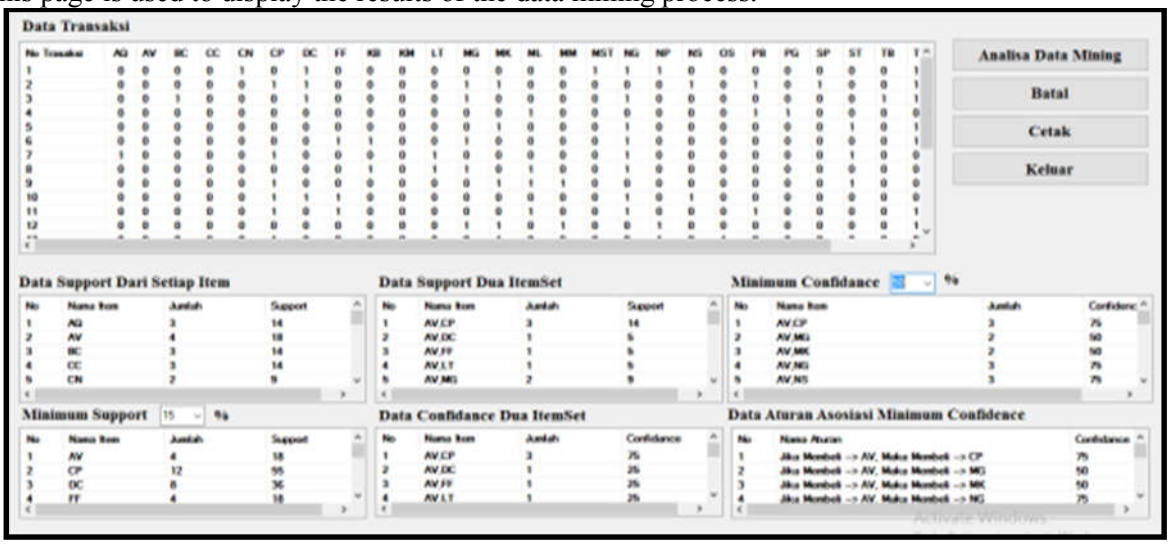

Figure2. Analysis Form

4. Conclusions

Based on the discussion and implementation, several conclusions can be obtained:

1. Analysis of consumer purchasing patterns using a priori algorithm in this application, can provide association rules by giving the drink value suport and drink confidence as a reference.
2. From the results of testing by entering a different value of support and drinking confidence, that is, if the minimum support is 15% and the confidence level is 50%, a rule of 87 rules will be generated.
3. The benefits of the test results for the cafe include:
   a. Fried Rice Stocks and Capuucino stock can be provided with many of the same stock.
   b. Cappuccino stock and stock of Kuah Noodle can be provided with many of the same stock.
   c. The tea stock pull and stock of fried noodles can be provided with many of the same stock
   d. The cafe can make a jumbo package food menu for more interesting

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